

Fig. 1.17 The standard rhombic lattice in the plane is shown at *left*; at *right* is a regular hexagonal lattice which can be viewed as a rhombic lattice with $n_x = n_y$ and $\theta = 120°$; it can also be viewed as a rhombic lattice with $\theta = 60°$

atoms i and j, and we wish to confine these to the cube $[0, L] \times [0, L] \times [0, L]$, then periodic boundary conditions involve an extended potential energy of the form

$$U^{\text{pbc}}(\boldsymbol{q}) = \sum_{klm} \sum_{i=1}^{\text{N}-1} \sum_{j=i+1}^{\text{N}} \varphi_{ij}(\boldsymbol{q}_i, \boldsymbol{q}_j + k\boldsymbol{v}_1 + l\boldsymbol{v}_2 + m\boldsymbol{v}_3),$$

where k, l, m run over $-1, 0, 1$ (in case interactions are restricted to the simulation cell and its immediate neighboring copies), and $\boldsymbol{v}_i^T = (L\mathbf{e}_i^T, L\mathbf{e}_i^T, \ldots, L\mathbf{e}_i^T)$, $i = 1, 2, 3$, where $\mathbf{e}_i$ is the ith Euclidean basis vector in $\mathbb{R}^3$. During simulation, the coordinates of each atom must be checked and possibly shifted after each positional step. The use of periodic boundary conditions may be likened to placing the molecular system (the configurational coordinates) on the surface of a torus. It may seem that periodic boundary conditions greatly increase the number of interactions that must be tracked in simulation, but this is not normally the case. First, for short-ranged interactions, such as those defined by the Lennard-Jones potential, the distant force terms will be negligible and can simply be taken to be zero. A *cut-off* is usually employed based on the interatomic distance. (See Exercise 11 for an illustration.) With or without periodic boundary conditions there will be only a finite number of close neighbors that actual contribute to the energy or force sum. The *minimum image convention* states that, in computing the force, a given atom interacts only with the nearest replica of any other atom.

The low potential energy configurations are natural starting points for simulation, and these normally correspond to a lattice structure. In 2D, the typical geometry observed at low temperature is defined by a rhombic lattice with sides of fixed length n_x, n_y and the angle between them (θ), see Fig. 1.17. For a system of uniform molecular pair potentials (such as Lennard-Jones), without boundary conditions, the equilibrium state will typically be close to the hexagonal lattice.

3D systems, particularly crystalline materials, favor lattice structures which can be viewed as stackings of planar layers. The *unit cell* is a description of the arrangement of atoms within a box; unit cells may be stacked in each direction to describe an atomic lattice. Three of the most common unit cells are the body-centered cubic (bcc), face-centered cubic (fcc) and hexagonal close-packed